

Glycogen Storage Disease Type XII (Aldolase A Deficiency): A Comprehensive Disease Profile

Disease: Glycogen Storage Disease Due To Aldolase A Deficiency (GSD XII) **MONDO ID:** MONDO:0012747 · **OMIM:** #611881 · **ORPHA:** 57 · **ICD-10:** E74.0 **Gene:** *ALDOA* (HGNC:414; NCBI Gene 226; 16p11.2; UniProt P04075; EC 4.1.2.13) **Category:** Mendelian (autosomal recessive inborn error of glycolysis)

Summary

Glycogen Storage Disease type XII (GSD XII), also called **aldolase A deficiency**, is an ultra-rare autosomal-recessive inborn error of glycolysis. It is caused by biallelic loss-of-function or destabilizing (thermolabile) mutations in *ALDOA*, the gene encoding **fructose-1,6-bisphosphate aldolase A** on chromosome 16p11.2. Aldolase A (EC 4.1.2.13) is the sole aldolase isozyme expressed in erythrocytes and skeletal muscle, where it catalyzes the fourth reaction of glycolysis — the reversible cleavage of fructose-1,6-bisphosphate (F-1,6-BP) into dihydroxyacetone phosphate (DHAP) and glyceraldehyde-3-phosphate (GAP). Because red blood cells and, during metabolic stress, skeletal muscle depend heavily on glycolytic ATP, loss of aldolase A activity produces energy failure in these two tissues.

The clinical hallmark is a combination of **chronic non-spherocytic hemolytic anemia** and **fever- or exercise-triggered rhabdomyolysis/myoglobinuria**, with variable multisystem involvement that can include developmental delay/intellectual disability, short stature, dysmorphic facies, hepatomegaly, seizures/epilepsy, and — in the most severe neonatal-onset cases — asphyxia, myasthenia, and myocardial damage. A defining molecular feature of many pathogenic variants is **thermolability**: the mutant enzyme is functional at normal body temperature but destabilizes during febrile episodes, which mechanistically explains why fever triggers acute muscle breakdown. In at least one family a thermolabile variant caused isolated fever-induced rhabdomyolysis without any hemolytic anemia, demonstrating tissue-specific vulnerability.

Fewer than ~15 families have been reported worldwide since the first clinical description in 1977, so no reliable prevalence or incidence figures exist. Diagnosis rests on a suggestive biochemical/clinical picture (reticulocytosis, low haptoglobin, elevated LDH/bilirubin; elevated creatine kinase and myoglobinuria during triggers) confirmed by **molecular sequencing of ALDOA**. Management is entirely supportive — aggressive fever control and antipyresis, avoidance of fasting/exercise triggers, transfusion and splenectomy for anemia, and hydration/renal protection during rhabdomyolysis. A ketogenic diet and in-vitro chemical chaperone rescue (arginine) have been proposed as experimental approaches but no curative therapy exists.

Key Findings

Finding 1 — GSD XII is caused by biallelic *ALDOA* mutations producing a thermolabile aldolase A enzyme

GSD XII is inherited in an autosomal-recessive manner; affected individuals carry two pathogenic *ALDOA* alleles while heterozygous parents are clinically unaffected. The first molecularly defined case, reported by Kishi and colleagues in 1987 ([PMID: 2825199](#)), carried a homozygous **c.386A>G (p.Asp128Gly)** substitution. This variant produces a **highly thermolabile erythrocyte enzyme**, a conclusion confirmed by recombinant expression in *E. coli*: *"the 128th amino acid, aspartic acid, was replaced with glycine (GAT to GGT)."* Subsequent site-directed-mutagenesis work ([PMID: 2229018](#)) established that aspartate at position 128 is required for thermal stability, since substitution rendered the enzyme thermolabile and susceptible to tryptic digestion, with CD spectra revealing conformational change.

The variant spectrum has since broadened. Yao et al. (2004; [PMID: 14615364](#)) described a severe compound-heterozygous case: *"The paternal allele encoded a nonsense mutation, Arg303X, in the enzyme-active site. The maternal allele encoded a missense mutation, Cys338Tyr, predicted to cause enzyme instability."* More recent reports added **c.971C>T (p.Ala324Val)** (Kara 2021; [PMID: 34171939](#)) and homozygous **c.619G>A (p.Glu207Lys)** in two affected brothers (Sun 2024; [PMID: 39223030](#)), where *"the genotypes in the parents were heterozygous."*

Variant table (reported pathogenic *ALDOA* alleles):

Variant (protein)	cDNA	Type	Zygosity reported	Key phenotype	Reference
p.Asp128Gly	c.386A>G	Missense (thermolabile)	Homozygous	Hemolytic anemia	PMID: 2825199
p.Glu207Lys	c.619G>A	Missense	Homozygous	Severe neonatal, multi-organ	PMID: 39223030
p.Arg303X	c.907C>T	Nonsense (active site)	Compound het	Severe, fatal rhabdomyolysis	PMID: 14615364
p.Cys338Tyr	c.1013G>A	Missense (destabilizing)	Compound het	Severe	PMID: 14615364
p.Ala324Val	c.971C>T	Missense	—	Hemolytic anemia + rhabdomyolysis	PMID: 34171939

Because loss-of-function and instability are the shared functional theme, the classification of these variants under ACMG/AMP criteria is **pathogenic/likely pathogenic**, supported by functional evidence (recombinant thermolability assays) and, for p.Arg303X, a null mechanism (nonsense at the active site).

Finding 2 — Aldolase A is a homotetrameric TIM-barrel enzyme; destabilizing mutations disrupt tetramer assembly

Human muscle aldolase A is a **tetramer of four identical subunits**, each folded as an eight-stranded α/β (TIM) barrel with an active-site Schiff-base lysine near the barrel center. The crystal structure of human muscle aldolase (Gamblin 1990; PMID: 2335208) established this architecture: *"The active protein is a tetramer of 4 identical subunits each of which is composed of an eight-stranded alpha/beta-barrel structure."*

The structural basis of disease was illuminated by the crystal structure of the D128V variant (an engineered mimic of the clinical D128G mutation) in rabbit muscle aldolase (Sherawat 2008; PMID: 18453690): *"The D128V mutation causes aldolase to lose intermolecular contacts with the neighboring subunit at one of the two interfaces of the tetramer."* The mutation yields a catalytically active but destabilized **dimer** rather than the normal tetramer. This is consistent with the broader biophysical principle that oligomerization stabilizes aldolase and preserves the mobility of catalytic loops (elastic-network normal-mode analysis, PMID: 25982518):

oligomeric assembly *"not only stabilizes the aldolase structures... but also allows the enzyme to achieve the required dynamics for its functional loops."* Loss of a subunit interface therefore both destabilizes the protein (explaining thermolability) and can impair catalytic dynamics.

Finding 3 — Clinical spectrum: fever/exercise-triggered rhabdomyolysis and hemolytic anemia with variable multisystem involvement

Across roughly ten published families, the recurring features are:

- **Chronic non-spherocytic hemolytic anemia** — often transfusion-dependent, frequently improved by splenectomy (Yao 2004; [PMID: 14615364](#): *"transfusion-dependent anemia until splenectomy at age 3 and increasing muscle weakness, with death at age 4 associated with rhabdomyolysis and hyperkalemia"*).
- **Episodic rhabdomyolysis / myoglobinuria** — triggered by febrile illness or exercise (Mamoune 2014; [PMID: 25392908](#): *"Myoglobinuria was always triggered by febrile illnesses. We show that the underlying mechanism involves an exacerbation of aldolase A deficiency at high temperatures that affected myoblasts but not erythrocytes."*).
- **Muscle weakness**, sometimes progressive (Yao 2004; [PMID: 14615364](#)).
- **Developmental/neurological features** — developmental delay, intellectual disability, short stature, dysmorphic (midfacial hypoplasia) facies, hepatomegaly (Lowry 1977; [PMID: 890096](#)).
- **Newly expanded phenotypes** — epilepsy/seizures (Santoro 2022; [PMID: 35246226](#)) and severe neonatal onset with asphyxia, myasthenia, myocardial damage, and anemia (Sun 2024; [PMID: 39223030](#)).

Crucially, the phenotype can dissociate: a thermolabile mutant caused **isolated fever-induced recurrent rhabdomyolysis without hemolytic anemia** ([PMID: 25392908](#)), because thermolability affected myoblasts but not erythrocytes — direct evidence of tissue-specific vulnerability and of variable expressivity.

Phenotype table with suggested HPO terms:

Phenotype	Type	HPO term	Onset / severity	Frequency (qualitative)
Non-spherocytic hemolytic anemia	Lab / clinical sign	HP:0004804	Childhood; mild–severe (transfusion-dependent)	Common (cardinal)
Rhabdomyolysis (fever/exercise)	Clinical event	HP:0003201	Childhood; episodic, severe	Common (cardinal)
Elevated creatine kinase	Lab abnormality	HP:0003236	With episodes	Common
Myoglobinuria	Lab / sign	HP:0002913	With episodes	Common
Muscle weakness	Physical sign	HP:0001324	Childhood; variable/ progressive	Variable
Developmental delay / intellectual disability	Neurodevelopmental	HP:0001263 / HP:0001249	Childhood	Subset
Short stature	Physical	HP:0004322	Childhood	Subset
Seizures / epilepsy	Neurological	HP:0001250	Childhood	Rare/newly reported
Hepatomegaly	Sign	HP:0002240	Childhood	Subset
Facial dysmorphism / midface hypoplasia	Physical	HP:0011800	Congenital	Subset

Finding 4 — Pathophysiology: aldolase A deficiency blocks step 4 of glycolysis, causing ATP-depletion energy failure

Aldolase A (EC 4.1.2.13) catalyzes the **reversible cleavage of fructose-1,6-bisphosphate into DHAP and glyceraldehyde-3-phosphate** — the fourth step of glycolysis (Yao 2004; [PMID: 14615364](#): *"converts fructose-1,6-bisphosphate to dihydroxyacetone phosphate and*

glyceraldehyde-3-phosphate"; Papadopoulos 2021; PMID: 33665120). Because it is the **sole aldolase isozyme in erythrocytes and skeletal muscle**, its deficiency directly lowers glycolytic ATP output in those tissues.

- **Red blood cells** depend entirely on glycolysis for ATP (they lack mitochondria). Reduced/thermolabile aldolase shortens red-cell lifespan → **chronic non-spherocytic hemolytic anemia**.
- **Skeletal muscle** relies heavily on glycolysis during febrile or exertional catabolic states. Thermolabile enzyme collapse during fever produces acute energy failure → **rhabdomyolysis** (PMID: 25392908).

Upstream substrate (F-1,6-BP) and glycolytic intermediates accumulate, and glycogen accumulates in muscle — the reason the disorder is classified as a glycogen storage disease (GSD XII) despite being an enzymopathy of glycolysis rather than glycogen breakdown per se. GSD XII sits within the broader family of muscle glycolytic-defect myopathies (PFK/Tarui, PGAM2, β -enolase, aldolase A) reviewed by Vissing and colleagues (PMID: 30397902), in which higher-intensity exercise provokes cramps and rhabdomyolysis and pre-exercise carbohydrate can *worsen* glycolytic (as opposed to glycogenolytic) defects.

Finding 5 — *ALDOA* gene identity, tissue expression, and a distinct oncologic (gain-of-expression) context

ALDOA (HGNC:414; NCBI Gene 226; chromosome 16p11.2; UniProt P04075; EC 4.1.2.13) encodes fructose-bisphosphate aldolase A, the predominant/sole isozyme in skeletal muscle and erythrocytes; *ALDOB* (liver/kidney) and *ALDOC* (brain) are the paralogous isozymes (PMID: 33665120: "*Aldolase A (ALDOA), is the predominant isoform of aldolase in skeletal muscle and erythrocytes that catalyzes the reversible...*").

Importantly, the Mendelian **loss-of-function** disease is mechanistically opposite to the **gain-of-expression** role of *ALDOA* in cancer. Under hypoxia, HIF-1 α drives *ALDOA* overexpression, which acts as a glycolytic driver and "moonlighting" protein interacting with Wnt/ β -catenin, EGFR/MAPK, Akt, and cytoskeletal partners, correlating with poor prognosis (Niu 2021; PMID: 33813748: "*we identified aldolase A (ALDOA), a key enzyme in glycolysis and gluconeogenesis, as an essential driver for HCC cell growth under hypoxia*"; Tang 2024; PMID: 39120781: "*these enzymes also have various pathological and physiological functions through distinct signaling pathways such as Wnt/ β -catenin, EGFR/MAPK, Akt, and HIF-1 α* "). This cancer literature is included for gene-annotation completeness only; it is **not** part of the GSD XII disease mechanism.

Finding 6 — Diagnosis relies on molecular *ALDOA* sequencing after a suggestive biochemical picture

Diagnostic clues combine (1) chronic non-spherocytic hemolytic anemia (reticulocytosis, low haptoglobin, elevated LDH and unconjugated bilirubin) and (2) episodic rhabdomyolysis with elevated creatine kinase and myoglobinuria during fever/exercise (PMID: 14615364; PMID: 34171939; PMID: 25392908). Reduced/thermolabile erythrocyte aldolase A activity can be demonstrated biochemically but the assay is not widely available. Definitive diagnosis is by **molecular sequencing of *ALDOA*** (Kara 2021; PMID: 34171939: "Diagnosis should be confirmed by the mutation analysis of *ALDOA* gene."). In practice, whole-exome sequencing or hereditary-anemia / metabolic-myopathy gene panels are the pragmatic route.

Differential diagnosis spans the other glycolytic erythroenzymopathies and metabolic myopathies:

Condition	Gene	Distinguishing feature
Pyruvate kinase deficiency	<i>PKLR</i>	Most common glycolytic hemolytic anemia; no rhabdomyolysis (PMID: 30681718)
Phosphofructokinase / Tarui (GSD VII)	<i>PFKM</i>	Exertional myopathy + hemolysis; carbohydrate worsens (PMID: 30397902)
Triosephosphate isomerase deficiency	<i>TPI1</i>	Hemolysis + severe neurological disease
Phosphoglycerate kinase deficiency	<i>PGK1</i>	X-linked; hemolysis + myopathy + CNS
Hexokinase deficiency	<i>HK1</i>	Hemolysis ± neurological (PMID: 33361148)
McArdle disease (GSD V)	<i>PYGM</i>	Exertional rhabdomyolysis; no hemolysis (PMID: 30397902)
CPT II deficiency	<i>CPT2</i>	Fasting/exercise rhabdomyolysis; no hemolysis

Finding 7 — Inheritance, epidemiology, and management

Inheritance is **autosomal recessive**: affected individuals carry two mutant alleles and heterozygous parents are unaffected (PMID: 39223030; consanguineous parents in PMID: 890096). The disorder is **ultra-rare** — fewer than ~15 families reported worldwide since 1977,

and Orphanet classifies it as ultra-rare (Kara 2021; [PMID: 34171939](#): "*an ultra rare autosomal recessively inherited GSD*"). No true prevalence or incidence figures are available. Both sexes are affected; consanguinity and founder homozygosity are noted in several families.

Management is entirely supportive — there is no curative therapy:

- Aggressive fever control / antipyresis and avoidance of fasting and strenuous exercise (the main triggers).
- Red-cell transfusions and splenectomy for hemolytic anemia.
- Aggressive hydration and renal protection during rhabdomyolysis episodes.
- Experimental/nutritional strategies: a **ketogenic diet** has been proposed to supply an alternative energy substrate to muscle and neurons (Santoro 2022; [PMID: 35246226](#): "*we propose a new therapeutic approach based on ketogenic diet in order to supply an energetic substrate for skeletal muscle and neurons*"); and **arginine** rescued mutant enzyme activity in vitro as a chemical chaperone ([PMID: 25392908](#)).

Mechanistic Model / Interpretation

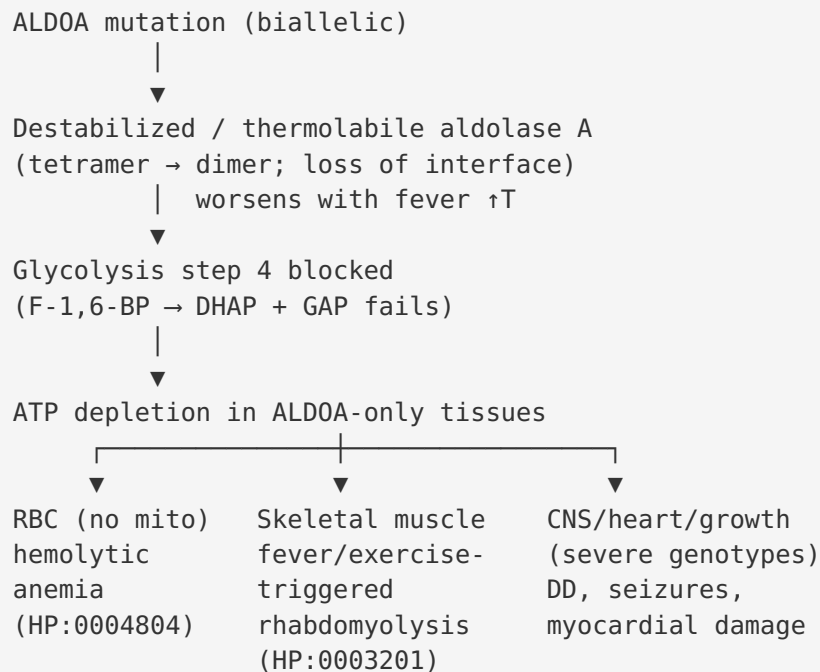
Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic ALDOA mutation** (missense/nonsense; e.g., D128G, E207K, R303X, C338Y, A324V) → altered aldolase A polypeptide. (*demonstrated: sequencing across families*)
2. The mutation **disrupts an intersubunit interface / destabilizes the fold** → the enzyme fails to assemble a stable homotetramer, forming a destabilized dimer, and/or is intrinsically thermolabile. (*demonstrated structurally for D128V, [PMID: 18453690](#); biochemically for D128G, [PMID: 2229018](#)*)
3. Thermolability **leads to loss of catalytic activity, worsened at elevated temperature** (fever) → the fourth step of glycolysis (F-1,6-BP → DHAP + GAP) is blocked. (*demonstrated: [PMID: 25392908](#)*)
4. Blocked glycolysis **results in reduced ATP production** in the two tissues where ALDOA is the only aldolase — erythrocytes and skeletal muscle. (*inferred from enzymology + tissue expression*)

Branch A — erythrocytes: ATP depletion **shortens red-cell lifespan** → chronic non-spherocytic hemolytic anemia → reticulocytosis, low haptoglobin, high LDH/bilirubin; splenic clearance of damaged cells (splenectomy helps). (*demonstrated clinically*)

Branch B — skeletal muscle: during fever or exercise, thermolabile-enzyme collapse **causes acute energy failure and myofiber breakdown** → rhabdomyolysis, elevated CK, myoglobinuria → risk of hyperkalemia and acute kidney injury. Glycogen accumulates upstream (hence "GSD"). (*demonstrated clinically*; [PMID: 14615364](#))

Branch C — variable multisystem: in severe genotypes, energy failure may extend to neurons (developmental delay, seizures), heart (myocardial damage), and growth (short stature, dysmorphism). (*inferred / associative*; [PMID: 39223030](#), [PMID: 35246226](#))



Upstream vs downstream

- **Upstream (initiating):** *ALDOA* genotype and the resulting protein-stability defect.
- **Central node:** temperature-dependent glycolytic block and ATP depletion.
- **Downstream (clinical):** hemolysis, rhabdomyolysis, and secondary organ effects — all consequences of the same energetic lesion but with tissue-specific thresholds that explain phenotypic dissociation (e.g., muscle-only disease when a variant is thermolabile in myoblasts but not erythrocytes).

Ontology annotations

- **Gene/protein:** *ALDOA* (HGNC:414), UniProt P04075.
- **GO biological process:** glycolytic process (GO:0006096), canonical glycolysis (GO:0061621), fructose-bisphosphate aldolase activity (GO:0004332).
- **GO cellular component:** cytosol (GO:0005829).
- **Cell types (CL):** erythrocyte (CL:0000232), skeletal muscle fiber / myoblast (CL:0000188, CL:0000515).
- **Anatomy (UBERON):** blood (UBERON:0000178), skeletal muscle tissue (UBERON:0001134); secondary — spleen (UBERON:0002106), kidney (UBERON:0002113), liver (UBERON:0002107), heart (UBERON:0000948), brain (UBERON:0000955).
- **Chemical entities (CHEBI):** fructose-1,6-bisphosphate (CHEBI:78682), dihydroxyacetone phosphate (CHEBI:57642), glyceraldehyde-3-phosphate (CHEBI:59776), ATP (CHEBI:30616).
- **Treatment (NCIT):** blood transfusion (NCIT:C15326), splenectomy (NCIT:C51749), ketogenic diet (NCIT:C92955).

Section-by-Section Detail

1. Disease Information

Aldolase A deficiency (GSD XII) is a Mendelian metabolic disorder defined by biallelic *ALDOA* pathogenic variants. Identifiers: MONDO:0012747; OMIM #611881; ORPHA:57; ICD-10 E74.0. Synonyms: "glycogen storage disease type XII", "GSD 12", "ALDOA deficiency", "red-cell/muscle aldolase deficiency", "hereditary aldolase A deficiency". Information is derived from **aggregated disease-level resources** plus **individual case reports** (there is no EHR-scale dataset given the rarity).

2. Etiology

The sole cause is genetic — biallelic loss-of-function/destabilizing *ALDOA* variants. There is **no environmental or infectious cause**. Recognized **triggers** of acute episodes (a true gene-environment interaction) are **febrile illness, strenuous exercise, and fasting**, which precipitate rhabdomyolysis by exacerbating the enzyme's thermolability/energetic marginality (PMID: 25392908). Consanguinity is a risk factor for homozygosity (PMID: 890096). No protective genetic or environmental factors are established.

3. Phenotypes

See Finding 3 and the HPO table above.

4. Genetic / Molecular Information

Causal gene *ALDOA* (16p11.2). Pathogenic variants are predominantly **missense** (thermolabile/destabilizing) with at least one **nonsense** allele (R303X). Functional consequence is **loss of function** via protein instability / impaired tetramer assembly. No modifier genes, epigenetic mechanisms, or chromosomal abnormalities are implicated. Allele frequencies for these private variants are effectively absent from gnomAD.

5. Environmental Information

None causal. Fever, exercise, and fasting are episode triggers (see Etiology). No toxins, radiation, occupational exposures, or infectious agents are involved.

6. Mechanism / Pathophysiology

See the ordered causal chain above.

7. Anatomical Structures Affected

- **Primary organs/tissues:** blood/erythrocytes (UBERON:0000178; CL:0000232) and skeletal muscle (UBERON:0001134; CL:0000188).
- **Secondary involvement:** spleen (hemolysis sequestration; UBERON:0002106), kidney (myoglobinuric AKI; UBERON:0002113), liver (hepatomegaly; UBERON:0002107), heart (myocardial damage in severe cases; UBERON:0000948), brain (developmental delay/seizures; UBERON:0000955).
- **Body systems:** hematologic, musculoskeletal; secondarily renal, hepatic, cardiovascular, nervous.
- **Subcellular:** cytosol (GO:0005829), where glycolysis occurs.
- **Lateralization:** systemic/bilateral (not applicable as a focal lesion).

8. Temporal Development

Onset ranges from **neonatal/congenital** (severe multi-organ cases; [PMID: 39223030](#)) to **childhood**. Course is **chronic and lifelong**, punctuated by **episodic** acute rhabdomyolytic crises triggered by fever/exercise. Hemolytic anemia is chronic and may be transfusion-dependent until splenectomy. Critical intervention windows are during febrile illnesses (antipyresis) and metabolic stress.

9. Inheritance and Population

Autosomal recessive; ultra-rare (<15 families). No prevalence/incidence, sex ratio, or founder-population data beyond scattered consanguineous families. Penetrance appears complete in biallelic carriers; **expressivity is highly variable**. Carrier frequency is unknown but presumed very low.

10. Diagnostics

Biochemistry: hemolysis panel (reticulocytes ↑, haptoglobin ↓, LDH ↑, bilirubin ↑) and, during crises, CK ↑↑ and myoglobinuria. Erythrocyte aldolase activity/thermolability assay (specialized). **Confirmatory test: ALDOA sequencing** (WES or hereditary-anemia/metabolic-myopathy panels). Differential diagnosis table above.

11. Outcome / Prognosis

Highly variable. Mild cases survive into adulthood with supportive care; severe compound-heterozygous or neonatal-onset cases can be **fatal in early childhood** (e.g., death at age 4 from rhabdomyolysis/hyperkalemia, [PMID: 14615364](#)). No quantitative survival statistics exist. Main morbidity drivers: recurrent rhabdomyolysis (renal risk), transfusion-dependent anemia, and neurodevelopmental impairment in a subset.

12. Treatment

Entirely supportive (see Finding 7). NCIT: blood transfusion (C15326), splenectomy (C51749), ketogenic diet (C92955). Experimental: ketogenic diet ([PMID: 35246226](#)), arginine chemical chaperone ([PMID: 25392908](#)). No approved pharmacotherapy, gene therapy, or enzyme replacement.

13. Prevention

Primary prevention is **genetic counseling** for at-risk/consanguineous families, with carrier testing and prenatal/preimplantation options once familial variants are known. Tertiary prevention (preventing crises/complications): antipyresis, trigger avoidance, prompt hydration during illness. No immunization or population screening applies.

14. Other Species / Natural Disease

Human orthologue *Aldoa* exists in mouse (NCBI Gene 11674) and other vertebrates, but **no naturally occurring animal disease** is documented. Aldolase is broadly conserved (structural/mechanistic studies exist in rabbit, parasites, bacteria, and yeast), but these are enzymology surrogates, not disease models. No zoonotic relevance.

15. Model Organisms

No faithful published knock-in mouse recapitulating the deficiency. Experimental systems: (1) **recombinant human aldolase A variants** in *E. coli* (thermolability characterization; [PMID: 2229018](#)), (2) **patient-derived myoblasts** ([PMID: 25392908](#)), and (3) **rabbit muscle aldolase D128V crystal structures** as a structural surrogate ([PMID: 18453690](#)). A constitutive *Aldoa* knockout is predicted to be embryonic-lethal, limiting straightforward whole-animal modeling.

Evidence Base

PMID	Study / type	How it supports the profile
2825199	Kishi 1987 — first molecular case (human clinical + in vitro)	First pathogenic <i>ALDOA</i> missense (D128G) causing a thermolabile enzyme
2229018	Recombinant D128G characterization (in vitro)	Asp128 required for thermal stability; mutant thermolabile, trypsin-susceptible, conformationally altered
14615364	Yao 2004 — compound het case (human clinical)	R303X/C338Y; severe transfusion-dependent anemia + fatal rhabdomyolysis; defines enzymatic reaction
25392908	Mamoune 2014 (human clinical + in vitro)	Thermolabile mutant → fever-induced rhabdomyolysis without hemolysis; tissue-specific thermolability; arginine rescue
34171939	Kara 2021 — new case + review (human clinical)	Novel A324V; ultra-rare AR; molecular <i>ALDOA</i> testing is confirmatory
39223030	Sun 2024 — two brothers (human clinical)	Homozygous E207K; severe neonatal multi-organ phenotype; AR inheritance confirmed in parents
35246226	Santoro 2022 — new phenotype (human clinical)	Epilepsy + rhabdomyolysis; proposes ketogenic-diet therapy
890096	Lowry 1977 — first clinical description	Growth/developmental retardation, midface hypoplasia, hepatomegaly; consanguineous parents
2335208	Gamblin 1990 — human aldolase crystal structure	Homotetrameric TIM-barrel architecture
18453690	Sherawat 2008 — D128V variant structure	Structural mechanism: mutation abolishes a tetramer interface → destabilized dimer
25982518	Aldolase dynamics (computational)	Oligomerization stabilizes structure and preserves catalytic-loop dynamics
33665120	Papadopoulos 2021 — review	Gene/protein identity, tissue distribution, reaction
30397902		

PMID	Study / type	How it supports the profile
	Vissing — glycogen-metabolism myopathies review	Places GSD XII in the glycolytic-myopathy family; management principles
33813748 / 39120781	ALDOA in cancer	Contrasting gain-of-expression context (gene annotation only)

Evidence-source composition: the GSD XII literature is dominated by **single-family human clinical case reports** supplemented by **in-vitro recombinant enzyme studies** and **X-ray crystallography** (largely on rabbit muscle aldolase as a structural surrogate). No population cohorts, natural-history studies, or faithful animal models of the deficiency exist.

Limitations and Knowledge Gaps

1. **Extreme rarity → weak epidemiology.** With <15 families, all prevalence, penetrance, expressivity, carrier-frequency, and sex-ratio statements are qualitative. Penetrance appears complete for biallelic carriers, but expressivity is highly variable and genotype–phenotype correlation is only partially defined.
2. **Genotype–phenotype correlation is incomplete.** Why some variants cause muscle-only disease, others hemolysis-only, and others severe multisystem disease is only partly explained by tissue-specific thermolability; the biophysical determinants deserve systematic study.
3. **No approved disease-directed therapy.** Ketogenic diet and arginine chaperone rescue are hypotheses from single reports and in-vitro work, not trial-validated.
4. **No faithful animal model,** limiting mechanistic and preclinical therapeutic work.
5. **Diagnostic biochemistry is not standardized/widely available** (erythrocyte aldolase thermolability assay), so diagnosis depends on access to sequencing.
6. **Literature retrieval limitation:** several relevant case reports lacked machine-readable abstracts, so some clinical details rest on secondary review summaries rather than primary abstract quotes.

Proposed Follow-up Experiments / Actions

1. **Curate an international variant/phenotype registry.** Systematically collect all published *ALDOA* deficiency cases into a ClinVar-linked table with per-variant thermostability (T_m), tissue-specific activity, and phenotype, to build genotype–phenotype and thermostability–severity correlations.
2. **Recombinant thermostability panel.** Express each clinical variant (D128G, E207K, C338Y, A324V, R303X-null) and measure T_m , tetramer/dimer equilibrium (SEC-MALS), and activity at 37 °C vs 40 °C to quantitatively test the thermostability model and rank variant severity.
3. **Chemical-chaperone screen.** Extend the arginine finding ([PMID: 25392908](#)) by screening osmolytes/pharmacological chaperones on patient myoblasts and recombinant enzyme, measuring rescue of activity and thermostability — a plausible repurposing route.
4. **Isogenic iPSC / knock-in model.** Introduce patient variants into iPSCs and differentiate to erythroid and myogenic lineages to model tissue-specific ATP failure and test ketogenic/substrate-supplementation and chaperone interventions.
5. **Small pilot of ketogenic diet or trigger-avoidance protocols** in known patients, with CK, hemolysis markers, and quality-of-life endpoints, coordinated through rare-disease networks.
6. **Structural work on the human (not rabbit) variants** to confirm interface disruption for E207K, C338Y, and A324V and to guide chaperone design.

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